

Multimodal Fusion of CLIP and ELECTRA for Interpreting Implicit Self-Harm Signals in Social Media Memes

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Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

Keywords: keyword1, Keyword2, Keyword3, Keyword4

1 Introduction

Social media has become a dominant infrastructure for digital communication, enabling users to share information, emotions, opinions, and personal experiences through text, images, videos, and multimodal content. DataReportal reported that active social media user identities reached 5.04 billion at the beginning of 2024, equivalent to 62.3% of the world population, with the typical user spending around 2 hours and 23 minutes per day on social media platforms [1]. Within this environment, internet memes have become a widely circulated form of digital expression. Shifman conceptualizes internet memes as groups of digital items that share common characteristics of content, form, or stance, and are circulated, imitated, or transformed by

many users through the internet [2]. In practice, memes often combine visual and textual elements to communicate humor, criticism, emotion, or social commentary in a compact and highly shareable format.

Although memes are commonly associated with entertainment, their meaning is not always explicit or harmless. A meme can appear neutral when its image or text is interpreted separately, but the combination of both modalities may produce a different contextual meaning. This makes meme interpretation challenging because the intended message often depends on visual-textual interaction, cultural references, sarcasm, irony, or indirect expression. In the Hateful Memes Challenge, Kiela et al. demonstrated that multimodal memes can be intentionally constructed so that unimodal models struggle, while successful interpretation requires joint reasoning over image and text [3]. This finding highlights a broader problem in meme understanding: models that rely only on visual or textual information may fail to capture meanings that emerge from the interaction between modalities.

One sensitive category of online content that requires careful attention is self-harm-related content. Clinical and public health guidelines define self-harm as an intentional act of self-injury or self-poisoning, regardless of underlying intent, and emphasize the importance of assessment, prevention, and recurrence management [4]. In online environments, self-harm-related content may appear not only in explicit forms but also through indirect emotional expressions, symbolic references, or ambiguous multimodal signals. A systematic review by Marchant et al. found that internet use can have both harmful and supportive roles for young people, depending on the type of content, context of exposure, and the way online spaces are used [5]. Reports and public health resources also emphasize that adolescents are a vulnerable group in relation to mental health challenges and online exposure [6, 7]. Therefore, detecting self-harm signals in memes should be approached as an assistive screening task rather than as an automatic clinical judgment.

The multimodal nature of memes makes them particularly difficult to analyze using conventional unimodal approaches. Multimodal machine learning aims to process and relate information from multiple modalities, such as images, text, audio, and video, to obtain a more complete representation of complex data [8]. This is especially relevant for memes because visual and textual components may complement, contradict, or modify each other. Prior work on harmful meme detection has shown that harmful meanings are often embedded in multimodal and context-dependent forms. Sharma et al. reviewed the harmful meme detection literature and found that most studies focus on categories such as hate speech and offensive content, while other harmful categories, including self-harm-related memes, remain underexplored due to limited publicly available datasets [9]. This indicates an important research gap in developing multimodal methods for detecting implicit self-harm signals in meme-based social media content.

Several recent studies have advanced multimodal meme understanding. MOMENTA proposes a multimodal framework for detecting harmful memes and identifying their targets by combining global and local perspectives from visual and textual modalities [10]. Hate-CLIPper further shows that modeling cross-modal interactions

between CLIP-based image and text representations can improve hateful meme classification [11]. MemeCLIP also demonstrates the usefulness of CLIP representations for multimodal meme classification tasks involving text-embedded images [12]. However, these studies mainly address harmful, hateful, offensive, or socially targeted memes. The specific problem of self-harm meme detection remains less explored, particularly when the signal is implicit, ambiguous, or expressed through the interaction between image and text.

To address this gap, this study proposes a multimodal classification framework that integrates Contrastive Language-Image Pre-training (CLIP) and Efficiently Learning an Encoder that Classifies Token Replacements Accurately (ELECTRA). CLIP learns transferable visual representations from natural language supervision by training on large-scale image-text pairs and aligning visual and textual information in a shared representation space [13]. ELECTRA introduces a sample-efficient text pre-training objective called replaced token detection, where the model learns to distinguish original tokens from replaced tokens across the input sequence [14]. By combining CLIP as the image encoder and ELECTRA as the text encoder, this study aims to capture both visual context and textual meaning in social media memes.

The proposed model applies an intermediate fusion strategy, where visual and textual representations are combined at the feature level before classification. This strategy is suitable for meme analysis because it allows the model to learn semantic interactions between image and text rather than relying on only one modality or combining final predictions at the decision level. The study evaluates the proposed multimodal approach against unimodal image-only and text-only baselines, examines several fusion and loss strategies, and analyzes model limitations on ambiguous memes using human annotation.

2 Related Work

Recent work on online meme analysis has made it increasingly clear that memes are not well modeled as image-only or text-only artifacts. The SemEval-2020 Memotion task formalized meme understanding as a multimodal problem spanning sentiment, humor, sarcasm, offensiveness, and motivational cues, while the later MOOD/ALFRED line further emphasized that affective interpretation in memes depends on the interplay between visual and textual signals rather than either modality in isolation [15, 16]. This broader view is consistent with general multimodal machine learning, where fusion is treated as a core challenge involving representation, alignment, and cross-modal interaction rather than simple feature concatenation [8].

Within harmful meme detection, the Hateful Memes Challenge became a major benchmark by deliberately constructing “benign confounders,” making unimodal shortcuts unreliable and forcing models to reason jointly over image and text [3]. Subsequent work has extended this line in several ways. MOMENTA modeled harmful memes through local and global multimodal perspectives and target identification [10], while Hate-CLIPper showed that explicitly modeling cross-modal interactions on top of CLIP representations can substantially improve hateful meme classification [11]. More recently, MemeCLIP demonstrated that CLIP-derived representations can also remain

effective for broader downstream meme understanding tasks beyond a single harmful category [12]. However, as noted in a dedicated survey of harmful memes, the literature has concentrated heavily on hate, offensive content, propaganda, and abuse, whereas categories such as self-harm remain substantially underexplored, largely because of limited datasets and the difficulty of operationalizing harm in implicit multimodal content [9].

Research on self-harm and suicide-related content detection on social media has developed mostly along a different trajectory: it has been predominantly text-centric and often focused on posts, timelines, or user-level histories rather than image-text memes. For example, Metzler et al. formulated a machine-learning framework for classifying harmful and protective suicide-related Twitter content across multiple categories [17], and Gaur et al. modeled suicidality on Reddit using clinician-adjudicated labels grounded in the Columbia Suicide Severity Rating Scale, highlighting the importance of severity and temporality in risk assessment [18]. This line of work is valuable, but it does not directly solve the meme problem, where self-harm signals may be encoded through embedded text, symbolic imagery, dark humor, or disagreement between the visual and textual channels. Recent work on broader sensitive-content classification has started to include self-harm as one moderation category, yet it likewise shows that open moderation tools and generic off-the-shelf models still struggle to address this category robustly across diverse social media content [19].

From the modeling perspective, CLIP and BERT-family encoders provide a strong foundation for this problem. CLIP learns transferable image representations from large-scale natural-language supervision and aligns visual and textual information in a shared embedding space [13]. On the language side, BERT established the effectiveness of deeply bidirectional pre-training for downstream language understanding [20], while ELECTRA introduced a more sample-efficient replaced-token detection objective that can yield stronger representations under comparable compute budgets [14]. These developments make CLIP a natural image encoder and ELECTRA a practical text encoder for meme classification, especially when the goal is not only visual grounding but also robust handling of short, noisy, and context-dependent meme text. Nevertheless, prior meme-detection studies have more often relied on vision-language transformers, CLIP-only pipelines, or generalized hateful-content frameworks than on a targeted CLIP-ELECTRA design for implicit self-harm memes [10–14, 20].

Another important strand of literature concerns deployment and moderation practice. Content moderation research increasingly treats automated classifiers as part of a human-AI collaboration pipeline rather than as stand-alone decision makers. Lykouris and Weng formalized moderation as a learning-to-defer problem in which uncertain cases should be routed to human review, explicitly accounting for reviewer capacity and delayed feedback [21]. Complementing that perspective, Wang et al. showed that LLM-generated moderation explanations can be fluent and persuasive yet still risk inducing erroneous judgments, underscoring the need for careful human oversight when the stakes are high [22]. For self-harm meme detection, this point is particularly important: ambiguous or affectively loaded cases are precisely the ones where full automation is least defensible. Accordingly, the present work is best situated not as a

clinical diagnostic system, but as a multimodal assistive model for early screening or reviewer support in sensitive-content analysis [21, 22].

Taken together, the literature suggests a clear gap. Meme understanding studies confirm that image-text interaction is essential [3, 8, 10–12, 15, 16]; harmful meme studies show strong progress but remain concentrated on hate, offense, and propaganda [3, 9–12]; and self-harm detection studies provide useful risk-modeling insights but are still largely textual and non-memetic [17–19]. This leaves relatively limited work on detecting implicit self-harm signals in social media memes through multimodal fusion. The present study addresses that gap by combining CLIP-based visual representations with ELECTRA-based textual representations under an intermediate fusion architecture, while also treating ambiguous cases as a human-supervision problem rather than a purely automatic classification task [13, 14, 20–22].

3 Dataset and Annotation

The scarcity of publicly available datasets specifically designed for self-harm memes remains a major obstacle for multimodal harmful-content research [9]. To address this limitation, we constructed a dataset from complementary primary and secondary sources so that the resulting corpus would cover both naturally occurring memes and more controlled multimodal risk expressions. The data collection process combined self-harm-related content gathered from social media, synthetic self-harm memes designed to enrich the representation of rare risk patterns, and general meme content obtained from public meme resources. In the initial stage, the study incorporated primary and secondary data, including synthetic augmentation to strengthen the representation of the self-harm class. After quality control and multimodal eligibility filtering, the final corpus consisted of 2,178 memes.

The synthetic component was introduced to mitigate the limited availability of self-harm meme examples while preserving the multimodal nature of the task. Rather than assuming that publicly sourced self-harm-related text or danger-related imagery could be directly treated as gold-standard labels, all such instances were passed through the same annotation pipeline as the rest of the corpus. This design choice is methodologically important because meme meaning often emerges from the interaction between image and text, not from either modality in isolation. In addition, the public meme subset was treated as a candidate pool of predominantly non-self-harm content rather than as an automatically trusted negative class. This conservative treatment reduces the risk of building the dataset on unchecked prior assumptions and is consistent with weak-supervision principles, in which noisy or indirect labeling sources are aggregated and then resolved through validation [23].

Annotation was carried out using a literature-informed guideline designed to support consistent multimodal judgment. The guideline was structured around three complementary views of interpretation: visual cues, textual cues, and the relation between image and text. This relational perspective is particularly important for memes because self-harm signals may be expressed indirectly through symbolic imagery, dark humor, irony, or cross-modal contradiction. In operational terms, a meme was labeled as self-harm when the combined visual-textual evidence conveyed

direct or indirect indicators associated with self-harm risk; otherwise, it was labeled as non-self-harm. The annotation protocol was therefore designed to evaluate not only explicit references, but also implicit multimodal signals whose meaning only becomes visible when both modalities are read together. This emphasis on multimodal rationale is conceptually compatible with recent work that highlights the value of rationale-aware analysis in multimodal safety assessment [24].

To improve labeling consistency while reducing the burden of exhaustive manual annotation, the study used an ensemble pseudo-labeling strategy. Two vision-language models, GPT-4o mini and Qwen-VL-Flash, independently annotated each meme using the same annotation guideline. When both models produced the same label, that label was accepted; when the outputs disagreed, the sample was manually reviewed by the researcher to determine the final label. This procedure reflects a weakly supervised labeling workflow in which multiple imperfect labeling sources are combined and disagreements are adjudicated rather than ignored [23]. In the final annotation stage, the agreement rate between the two models reached 94.95%, while 5.05% of samples required manual adjudication. The final dataset contained 653 self-harm memes and 1,525 non-self-harm memes.

Because the final corpus remained imbalanced, with the non-self-harm class representing approximately 70% of the data, the class distribution was retained to preserve a more realistic approximation of naturally occurring online content rather than being artificially equalized at the dataset-construction stage. This decision makes the downstream classification task more difficult, but it also prevents the paper from overstating performance under an overly sanitized distribution. At the same time, several limitations should be acknowledged. First, part of the self-harm class was supported by synthetic meme construction, which is useful for coverage but cannot fully reproduce the evolving vernacular of real-world online communities. Second, the dataset is limited to English-language textual content and to the platforms and public sources sampled in this study. Third, although ensemble pseudo-labeling improved consistency, semantic ambiguity remains unavoidable for memes involving sarcasm, metaphor, dark humor, or conflicting visual-textual cues. For ethical reasons, explicit examples should be minimized in the paper and any illustrative cases should be anonymized or carefully paraphrased.

4 Results

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

5 This is an example for first level head—section head

5.1 This is an example for second level head—subsection head

5.1.1 This is an example for third level head—subsubsection head

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

6 Equations

Equations in L^AT_EX can either be inline or on-a-line by itself (“display equations”). For inline equations use the `$...$` commands. E.g.: The equation $H\psi = E\psi$ is written via the command `$H \psi = E \psi$`.

For display equations (with auto generated equation numbers) one can use the `equation` or `align` environments:

$$\|\tilde{X}(k)\|^2 \leq \frac{\sum_{i=1}^p \|\tilde{Y}_i(k)\|^2 + \sum_{j=1}^q \|\tilde{Z}_j(k)\|^2}{p+q}. \quad (1)$$

where,

$$\begin{aligned} D_\mu &= \partial_\mu - ig \frac{\lambda^a}{2} A_\mu^a \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc} A_\mu^b A_\nu^c \end{aligned} \quad (2)$$

Notice the use of `\nonumber` in the `align` environment at the end of each line, except the last, so as not to produce equation numbers on lines where no equation numbers are required. The `\label{}` command should only be used at the last line of an `align` environment where `\nonumber` is not used.

$$Y_\infty = \left(\frac{m}{\text{GeV}} \right)^{-3} \left[1 + \frac{3 \ln(m/\text{GeV})}{15} + \frac{\ln(c_2/5)}{15} \right] \quad (3)$$

The class file also supports the use of `\mathbb{}`, `\mathscr{}` and `\mathcal{}` commands. As such `\mathbb{R}`, `\mathscr{R}` and `\mathcal{R}` produces $\mathbb{R}$, $\mathscr{R}$ and $\mathcal{R}$ respectively (refer Subsubsection 5.1.1).

7 Tables

Tables can be inserted via the normal `table` and `tabular` environment. To put footnotes inside tables you should use `\footnotetext[...]` tag. The footnote appears just below the table itself (refer Tables 1 and 2). For the corresponding footnotemark use `\footnotemark[...]`

The input format for the above table is as follows:

```
\begin{table}[<placement-specifier>]
\caption{<table-caption>}\label{<table-label>}%
\begin{tabular}{@{}llll@{}}
\toprule
Column 1 & Column 2 & Column 3 & Column 4\\
\midrule
row 1 & data 1 & data 2 & data 3 \\
\end{tabular}
```

Table 1 Caption text

Column 1	Column 2	Column 3	Column 4
row 1	data 1	data 2	data 3
row 2	data 4	data 5 ¹	data 6
row 3	data 7	data 8	data 9 ²

Source: This is an example of table footnote. This is an example of table footnote.

¹Example for a first table footnote. This is an example of table footnote.

²Example for a second table footnote. This is an example of table footnote.

```
row 2 & data 4 & data 5\footnotemark[1] & data 6 \\
row 3 & data 7 & data 8 & data 9\footnotemark[2]\\
\botrule
\end{tabular}
\footnotetext{Source: This is an example of table footnote.
This is an example of table footnote.}
\footnotetext[1]{Example for a first table footnote.
This is an example of table footnote.}
\footnotetext[2]{Example for a second table footnote.
This is an example of table footnote.}
\end{table}
```

Table 2 Example of a lengthy table which is set to full textwidth

Project	Element 1 ¹			Element 2 ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 ± 12	780 A	1166	1239 ± 100
Element 4	500 A	961	922 ± 10	900 A	1268	1092 ± 40

Note: This is an example of table footnote. This is an example of table footnote this is an example of table footnote this is an example of table footnote.

¹Example for a first table footnote.

²Example for a second table footnote.

In case of double column layout, tables which do not fit in single column width should be set to full text width. For this, you need to use `\begin{table*} ... \end{table*}` instead of `\begin{table} ... \end{table}` environment. Lengthy tables which do not fit in textwidth should be set as rotated table. For this, you need to use `\begin{sidewaystable} ... \end{sidewaystable}` instead of `\begin{table*}`

... `\end{table*}` environment. This environment puts tables rotated to single column width. For tables rotated to double column width, use `\begin{sidewaystable*}` ... `\end{sidewaystable*}`.

8 Figures

As per the $\text{\LaTeX}$ standards you need to use eps images for $\text{\LaTeX}$ compilation and pdf/jpg/png images for PDF $\text{\LaTeX}$ compilation. This is one of the major difference between $\text{\LaTeX}$ and PDF $\text{\LaTeX}$. Each image should be from a single input .eps/vector image file. Avoid using subfigures. The command for inserting images for $\text{\LaTeX}$ and PDF $\text{\LaTeX}$ can be generalized. The package used to insert images in $\text{\LaTeX}$ /PDF $\text{\LaTeX}$ is the graphicx package. Figures can be inserted via the normal figure environment as shown in the below example:

```
\begin{figure}[<placement-specifier>]
\centering
\includegraphics{<eps-file>}
\caption{<figure-caption>}\label{<figure-label>}
\end{figure}
```



Fig. 1 This is a widefig. This is an example of long caption this is an example of long caption this is an example of long caption this is an example of long caption

In case of double column layout, the above format puts figure captions/images to single column width. To get spanned images, we need to provide `\begin{figure*}` ... `\end{figure*}`.

For sample purpose, we have included the width of images in the optional argument of `\includegraphics` tag. Please ignore this.

9 Algorithms, Program codes and Listings

Packages `algorithm`, `algorithmicx` and `algpseudocode` are used for setting algorithms in $\text{\LaTeX}$ using the format:

```
\begin{algorithm}
\caption{<alg-caption>}\label{<alg-label>}
```

Table 3 Tables which are too long to fit, should be written using the “sidewaystable” environment as shown here

Projectile	Element 1 ¹			Element ²		
	Energy	σ_{calc}	σ_{expt}	Energy	σ_{calc}	σ_{expt}
Element 3	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 4	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40
Element 5	990 A	1168	1547 $\pm$ 12	780 A	1166	1239 $\pm$ 100
Element 6	500 A	961	922 $\pm$ 10	900 A	1268	1092 $\pm$ 40

Note: This is an example of table footnote this is an example of table footnote this is an example of table footnote
this is an example of table footnote.

¹This is an example of table footnote.

```

\begin{algorithmic}[1]
. . .
\end{algorithmic}
\end{algorithm}

```

You may refer above listed package documentations for more details before setting `algorithm` environment. For program codes, the “verbatim” package is required and the command to be used is `\begin{verbatim} ... \end{verbatim}`.

Similarly, for listings, use the `listings` package. `\begin{lstlisting} ... \end{lstlisting}` is used to set environments similar to `verbatim` environment. Refer to the `lstlisting` package documentation for more details.

A fast exponentiation procedure:

```

begin
  for i:=1 to 10 step 1 do
    expt(2,i);
    newline() od
where
proc expt(x,n) ≡
  z:=1;
  do if n=0 then exit fi;
  do if odd(n) then exit fi;
    comment: This is a comment statement;
    n:=n/2; x:=x*x od;
  { n>0 };
  n:=n-1; z:=z*x od;
  print(z).
end

```

Comments will be set flush to the right margin

```

for i:=maxint to 0 do
begin
{ do nothing }
end;
Write('Case-insensitive-');
Write('Pascal-keywords. ');

```

10 Cross referencing

Environments such as `figure`, `table`, `equation` and `align` can have a label declared via the `\label{#label}` command. For figures and table environments use the `\label{}` command inside or just below the `\caption{}` command. You can then use the `\ref{#label}` command to cross-reference them. As an example, consider the label declared for Figure 1 which is `\label{fig1}`. To cross-reference it, use the command `Figure \ref{fig1}`, for which it comes up as “Figure 1”.

Algorithm 1 Calculate $y = x^n$

Require: $n \geq 0 \vee x \neq 0$ **Ensure:** $y = x^n$

```
1:  $y \leftarrow 1$ 
2: if  $n < 0$  then
3:    $X \leftarrow 1/x$ 
4:    $N \leftarrow -n$ 
5: else
6:    $X \leftarrow x$ 
7:    $N \leftarrow n$ 
8: end if
9: while  $N \neq 0$  do
10:  if  $N$  is even then
11:     $X \leftarrow X \times X$ 
12:     $N \leftarrow N/2$ 
13:  else [ $N$  is odd]
14:     $y \leftarrow y \times X$ 
15:     $N \leftarrow N - 1$ 
16:  end if
17: end while
```

To reference line numbers in an algorithm, consider the label declared for the line number 2 of Algorithm 1 is `\label{algln2}`. To cross-reference it, use the command `\ref{algln2}` for which it comes up as line 2 of Algorithm 1.

10.1 Details on reference citations

Standard L^AT_EX permits only numerical citations. To support both numerical and author-year citations this template uses `natbib` L^AT_EX package. For style guidance please refer to the template user manual.

Here is an example for `\cite{...}`: [1]. Another example for `\citep{...}`: [2]. For author-year citation mode, `\cite{...}` prints Jones et al. (1990) and `\citep{...}` prints (Jones et al., 1990).

All cited bib entries are printed at the end of this article: [3], [4], [5], [6], [7], [8], [9], [10], [11], [12] and [13].

11 Examples for theorem like environments

For theorem like environments, we require `amsthm` package. There are three types of predefined theorem styles exists—`thmstyleone`, `thmstyletwo` and `thmstylethree`

thmstyleone	Numbered, theorem head in bold font and theorem text in italic style
thmstyletwo	Numbered, theorem head in roman font and theorem text in italic style
thmstylethree	Numbered, theorem head in bold font and theorem text in roman style

For mathematics journals, theorem styles can be included as shown in the following examples:

Theorem 1 (Theorem subhead) *Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text. Example theorem text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proposition 2 *Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text. Example proposition text.*

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Example 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Remark 1 Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Definition 1 (Definition sub head) Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text. Example definition text.

Additionally a predefined “proof” environment is available: `\begin{proof} ... \end{proof}`. This prints a “Proof” head in italic font style and the “body text” in roman font style with an open square at the end of each proof environment.

Proof Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text. Sample body text.

Proof of Theorem 1 Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. Example for proof text. □

For a quote environment, use `\begin{quote}... \end{quote}`

Quoted text example. Aliquam porttitor quam a lacus. Praesent vel arcu ut tortor cursus volutpat. In vitae pede quis diam bibendum placerat. Fusce elementum convallis neque. Sed dolor orci, scelerisque ac, dapibus nec, ultricies ut, mi. Duis nec dui quis leo sagittis commodo.

Sample body text. Sample body text. Sample body text. Sample body text. Sample body text (refer Figure 1). Sample body text. Sample body text. Sample body text (refer Table 3).

12 Methods

Topical subheadings are allowed. Authors must ensure that their Methods section includes adequate experimental and characterization data necessary for others in the field to reproduce their work. Authors are encouraged to include RIIDs where appropriate.

Ethical approval declarations (only required where applicable) Any article reporting experiment/s carried out on (i) live vertebrate (or higher invertebrates), (ii) humans or (iii) human samples must include an unambiguous statement within the methods section that meets the following requirements:

1. Approval: a statement which confirms that all experimental protocols were approved by a named institutional and/or licensing committee. Please identify the approving body in the methods section
2. Accordance: a statement explicitly saying that the methods were carried out in accordance with the relevant guidelines and regulations
3. Informed consent (for experiments involving humans or human tissue samples): include a statement confirming that informed consent was obtained from all participants and/or their legal guardian/s

If your manuscript includes potentially identifying patient/participant information, or if it describes human transplantation research, or if it reports results of a clinical trial then additional information will be required. Please visit (<https://www.nature.com/nature-research/editorial-policies>) for Nature Portfolio journals, (<https://www.springer.com/gp/authors-editors/journal-author/journal-author-helpdesk/publishing-ethics/14214>) for Springer Nature journals, or (<https://www.biomedcentral.com/getpublished/editorial-policies#ethics+and+consent>) for BMC.

13 Discussion

Discussions should be brief and focused. In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Some journals prefer a section ‘Results and Discussion’ followed by a section ‘Conclusion’. Please refer to Journal-level guidance for any specific requirements.

14 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Please refer to Journal-level guidance for any specific requirements.

Supplementary information. If your article has accompanying supplementary file/s please state so here.

Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

Please refer to Journal-level guidance for any specific requirements.

Acknowledgements. Acknowledgements are not compulsory. Where included they should be brief. Grant or contribution numbers may be acknowledged.

Please refer to Journal-level guidance for any specific requirements.

Declarations

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- Funding
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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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